Exercise 1: Introduction (Projects & Git)

Software Architecture
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Organizational Matters

Lectures

- Tuesdays and Wednesdays (updated lecture schedules on Moodle)
- Contents, discussions, interactive formats
- Materials via Moodle

Written Exam

- Digital via Dynexite
- 2/3 of grade
- Facts, use cases, discussions, scenarios, trade-offs, diagrams, code
- Group projects (exercise) must be passed

Organizational Matters

Exercises

- Thursdays, based on previous lecture contents
- Contents, technical tutorials, labs with code examples
- Materials via Moodle, GitLab
- Tasks for each week for group projects in exercise sheet
- Time for on-site group discussions/work and teacher feedback (prepare questions, slides, demos)
- Note: Exercises are not mean to be debugging sessions with the teachers.

Group Projects (Homework)

- 1/3 of grade (need to be passed to take exam)
- Develop a software system (open domains), incrementally
- Groups of 4 – you stay in one group the entire semester
- Intro/overview and group forming in first exercise (Thursday, Oct 15)
- Having doubts about the course/project? Please deregister asap (others will be grateful).

Group Projects

- Topic/Domain: Free choice
- Programming Languages:
  - Core domain services in Java (Spring Boot), instructions on exercise sheets
  - Free choice for supporting services
- RWTH GitLab (NRW)
- Docker for local operation, VMs for hosting on demand via SWA

Today

- Example projects
- Small Git introduction

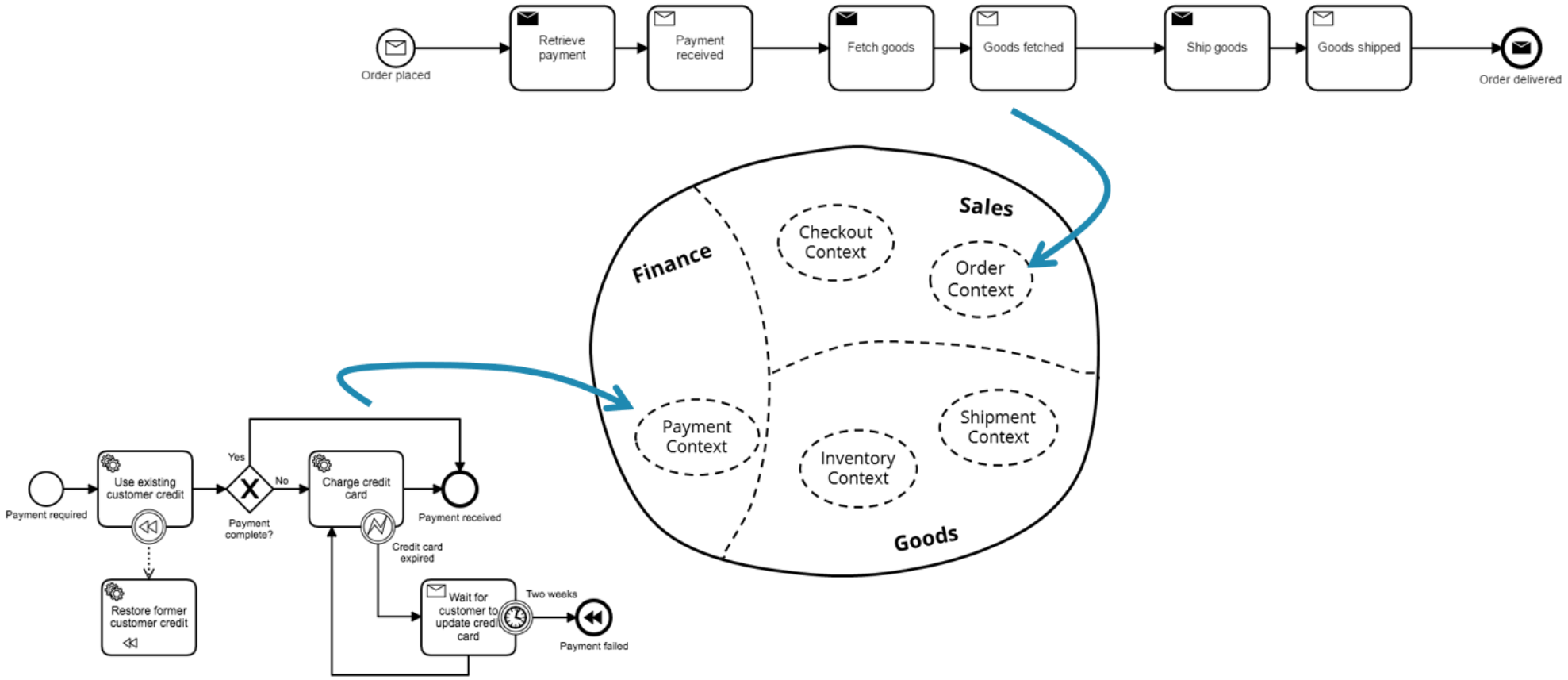
Projects

- Goal: workflows in a specific domain, involving multiple software components
- Start small, add components (services) with different features/responsibilities
- Two simple services in the first weeks
- Two more complex, structured services afterwards
- Later, a service that invokes an external LLM via RWTH KI:connect
- Services will interact (different modes will be investigated).
- Services will persist data in databases.
- Workflows (business processes) will be added on top, orchestrating services and potentially also human tasks

- Projects will grow incrementally.
- Exercise sheets will specify what you will have to do for the specific week.
  Do not start the implementation now! Wait for instructions.

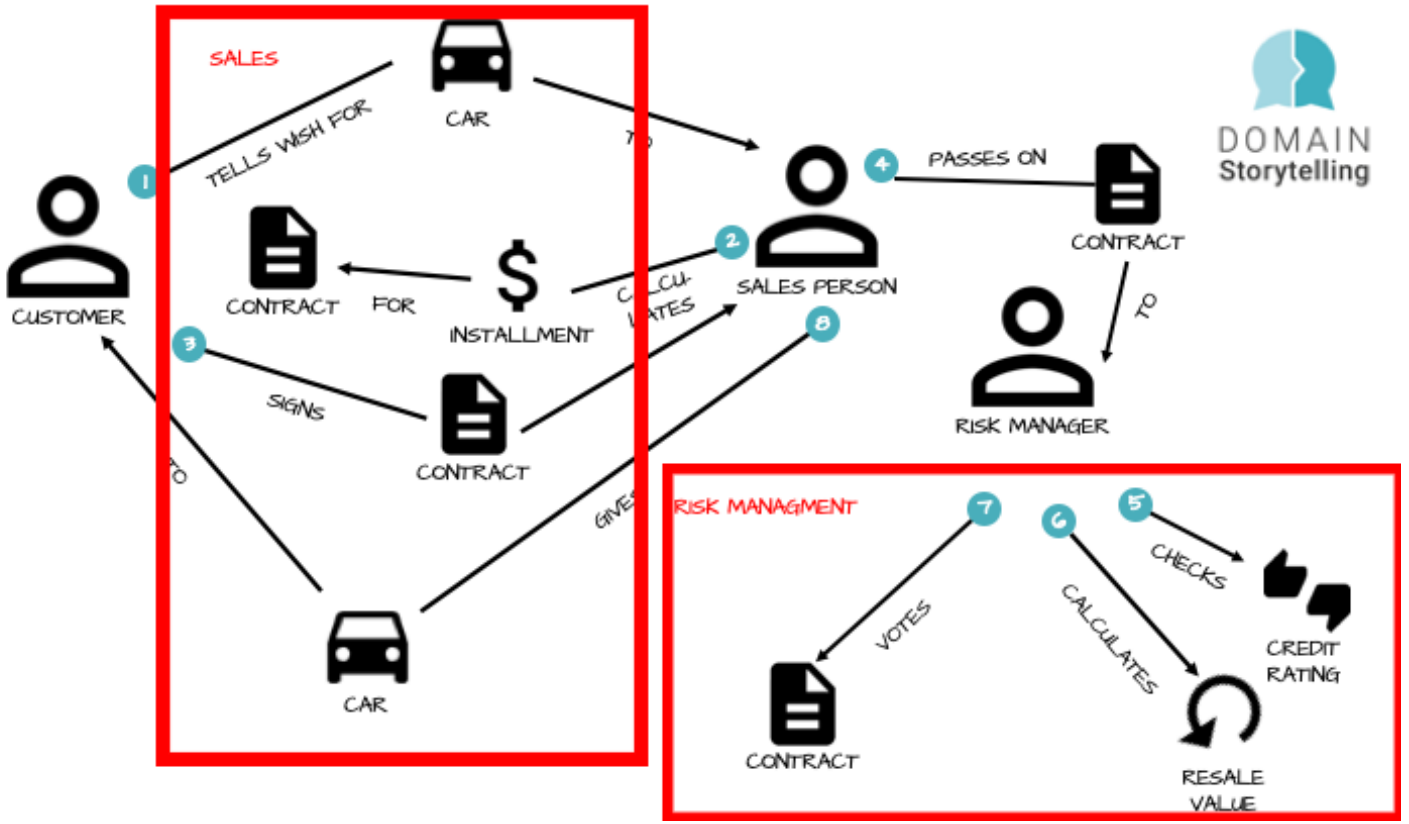
Example Projects

Flowing Retail: https://github.com/berndruecker/flowing-retail

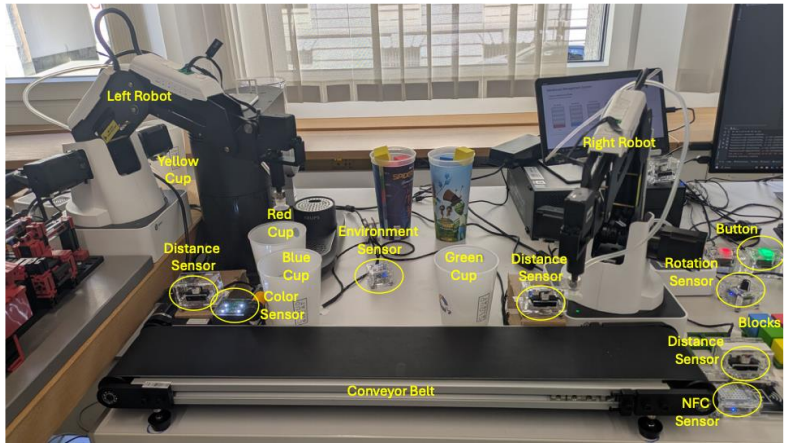


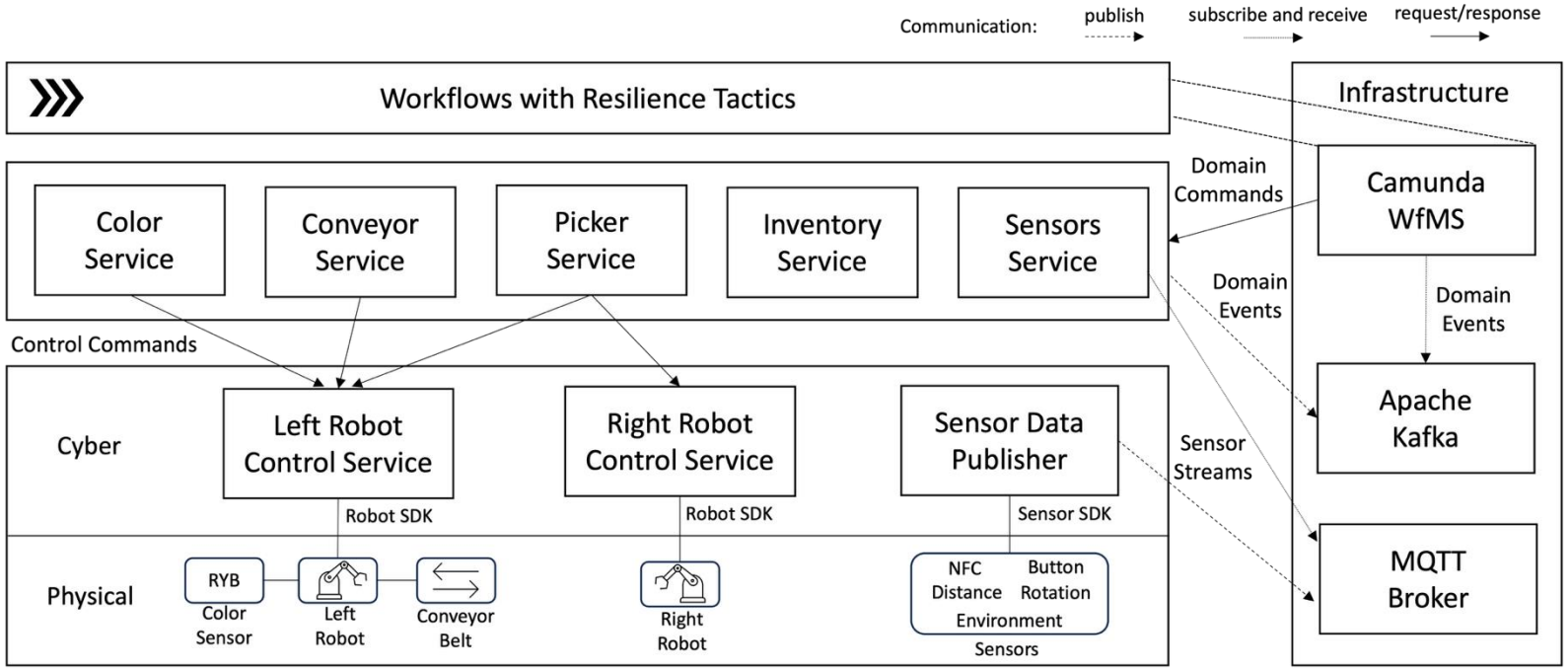
Example Projects

Leasing Ninja: https://leasingninja.io/



Example Projects (former student project at UniSG)





Example Projects (former student project at UniSG)



Git

Git

What?

- Version Control System (VCS)
- Created by Linux Torvalds in 2005



Why?

- Easier collaboration
  - no overwriting of each other‘s work
- included versioning
- complete change history
  - easy to go back

Git

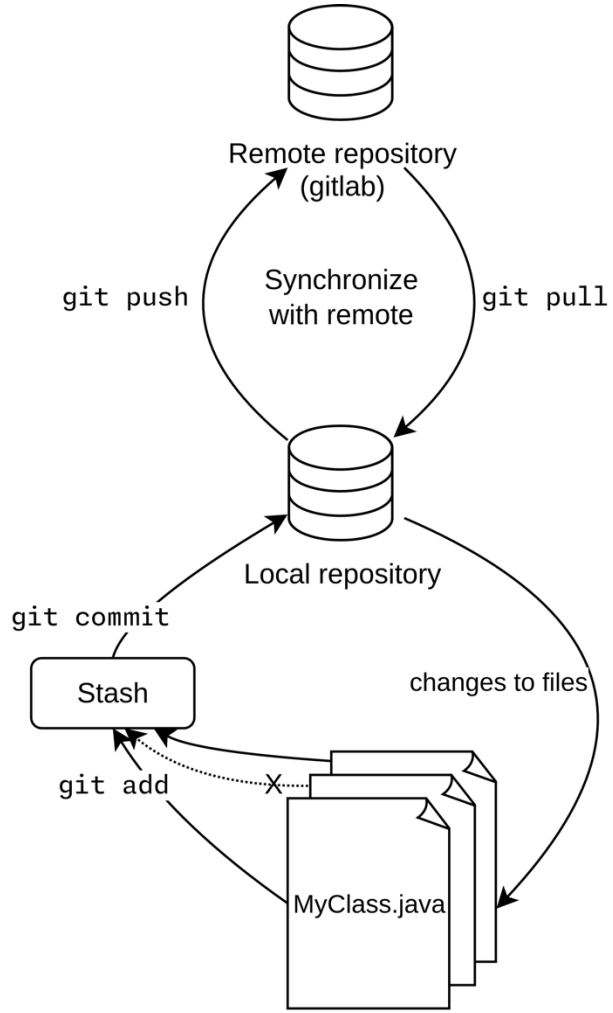
Basics

- *Repository:* folder tracked by git
- *Commit:* snapshot of the repository with changes
  - identified by a unique hash
- *Branch:* separate line of work
  - often: main, dev and feature-branches
- *Merge:* combining changes from different branches

Typical Workflow

- *Clone* the repository (initially)
- *Pull* the current version from the remote
- make changes, *add* and *commit* them locally
- *Push* changes to the remote repository
- Eventually create a *merge request*

Note: Code on *main* branch should always build and run.



Git

Common Problems

- Merge Conflicts
  - two commits alter the same file
  - sometimes, manual intervention required
- Unnecessary (junk) files under version control. Define .gitignore files!
- Large binary (non-text) files under version control. Every change and commit pollutes the repository.
- Once something is committed, it is not easy to remove it again (Git is designed to track and log all changes). Beware of sensible data, e.g., passwords, API keys, etc!

Recommendation

- Commit often
- feature-based branches
- communication!
- Leverage IDE integrations (e.g., in IntelliJ)

Exercise Tasks

- Network and find groups
- Signup for the groups in Moodle
- Deregister if you cannot/do not want to participate.

- Brainstorm project ideas
- Check exercise sheet on Moodle
- Create and configure Git repositories on GitLab (no GitHub!)

- Questions? Doubts? Concerns? Ask teachers!